

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

Course Code:	CSA1009	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
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Date:	06/10/2026	Duration:	60 minutes

I **G Rahul (192572008)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A –

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.

Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	Mark
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25

SMART CLASSROOM MANAGEMENT SYSTEM

Software Engineering – CO1 Case Study Analysis

Problem Statement: Develop a system for managing classroom schedules, course materials, assignments, announcements, student participation, and faculty activities.

1. Understand the Problem Statement

1.1 Problem Explanation

A Smart Classroom Management System is a centralized software solution for managing academic and classroom activities. In a traditional environment, schedules, course materials, assignments, announcements, participation records, and faculty activities may be handled through notice boards, spreadsheets, paper records, email, or different messaging and learning platforms. This creates scattered information and increases manual effort.

The proposed system brings these activities together in one platform. Users receive access according to their role, allowing students to view academic information and submit work, faculty to manage teaching activities, and administrators or department authorities to monitor and control the system. The main purpose is to make classroom management organized, accessible, secure, and efficient.

1.2 Objectives

1. Centralize classroom schedules, course materials, assignments, announcements, participation and faculty activities.
2. Reduce manual record keeping and paperwork.
3. Improve communication between faculty, students and administrators.
4. Provide timely access to academic information and notifications.
5. Simplify assignment creation, submission and tracking.
6. Maintain accurate participation and faculty activity records.
7. Provide role-based access and protect academic information.
8. Generate useful reports for monitoring and decision-making.

1.3 Scope

The system covers student, faculty and administrator functions within a department or college. It includes authentication, user management, timetable and classroom scheduling, course materials, assignments, announcements, student participation, faculty activities, dashboards and reports. The initial system can serve one department and later scale to multiple departments and the entire institution.

1.4 Assumptions and Constraints

- Users have valid institutional accounts and basic digital literacy.
- The college provides suitable network/server infrastructure.
- Academic data entered by authorized users is accurate.
- The system must follow institutional privacy and access policies.
- Availability of reliable internet/network access may be a deployment constraint.

2. Key Stakeholders and Challenges

2.1 Stakeholder Analysis

Stakeholder	Responsibilities	Main Needs / Challenges
Students	View schedules/materials, receive announcements, submit assignments and view participation.	Easy access, timely notifications, simple submission process and reliable records.
Faculty	Manage courses/materials, assignments, announcements, participation and activities.	Less manual work, quick updates, submission tracking and accurate records.
Administrator	Manage users, roles, courses, classrooms and system settings.	Central control, data accuracy, security and reporting.
HOD / Department	Monitor courses, faculty activities, schedules and student participation.	Department-level visibility and useful reports.
College Management	Review overall academic activities and performance information.	Reliable summaries for planning and decisions.
IT / System Administrator	Maintain application, database, backups, security and technical support.	System availability, security, recovery and maintainability.

2.2 Major Challenges

- Information is distributed across multiple channels.
- Manual timetable and classroom allocation can create conflicts.
- Assignment collection and tracking require repeated manual work.
- Students may miss important announcements or deadline changes.
- Participation records can be difficult to maintain consistently.
- Unauthorized access can expose academic and personal information.
- Network or server failures can interrupt access.
- Users may initially resist changing from familiar manual processes.

3. Analyze the Current Approach

3.1 Existing Process

In the current/traditional approach, the timetable may be displayed physically or shared as a file, materials may be distributed through email or messaging applications, assignments may be collected through paper or separate online tools, announcements may be communicated verbally or through messages, and participation may be maintained in registers or spreadsheets. Faculty activities can also be maintained separately.

3.2 Strengths

- Simple to begin with and requires little initial software infrastructure.
- Direct faculty–student communication is possible.
- Suitable for very small classes with limited data.
- Paper or spreadsheet records can be retained as evidence.

3.3 Limitations and Gaps

- No single source of truth for academic information.

- Manual updates are time-consuming and error-prone.
- Information can be missed when multiple communication channels are used.
- Searching and retrieving old records is difficult.
- Assignment status and participation are harder to monitor.
- Reports require additional manual effort.
- Access control and security may be inconsistent.

3.4 Need for the Proposed System

The major gap is the lack of an integrated platform connecting classroom management, teaching resources, assignments, communication, participation and faculty activities. A centralized system can reduce duplication, improve visibility, and make academic processes easier to monitor.

4. Proposed Solution Using Software Engineering Principles

4.1 System Overview

The proposed system will use a centralized application and database. After authentication, the system displays role-specific dashboards. Students can access schedules, materials, assignments and announcements. Faculty can manage course content, assignments, announcements, participation and activities. Administrators manage users, courses, classrooms and system settings. HODs and management can access authorized reports.

4.2 Major Functional Modules

1. Authentication and User Management

Registration/account creation, login, logout, password management, role assignment and access control.

2. Classroom and Schedule Management

Create and update timetables, allocate classrooms, manage faculty schedules and identify scheduling conflicts.

3. Course and Material Management

Create courses, upload notes/presentations/documents, organize materials and provide controlled student access.

4. Assignment Management

Create assignments, set deadlines, publish instructions, accept submissions, track status and support faculty review.

5. Announcement and Notification Management

Publish academic notices, schedule changes, deadline reminders and important updates.

6. Student Participation Management

Record classroom participation, maintain student records and provide participation summaries.

7. Faculty Activity Management

Maintain faculty course responsibilities, academic activities, schedules and related records.

8. Dashboard and Reporting

Provide role-based dashboards and reports for schedules, assignments, participation, courses and faculty activities.

4.3 Functional Requirements

9. The system shall authenticate users and provide role-based access.
10. The administrator shall manage users, courses, classrooms and system settings.
11. Faculty shall create and update course materials.
12. Students shall view and download authorized course materials.
13. Faculty shall create assignments and deadlines.
14. Students shall submit assignments and view submission status.
15. Faculty shall view and manage student submissions.
16. Authorized users shall create and view announcements.
17. Faculty shall record student participation.
18. Students shall view their own authorized participation information.
19. The system shall maintain faculty activity records.
20. The system shall provide dashboards and reports for authorized users.

4.4 Non-Functional Requirements

Requirement	How it is addressed
Usability	Simple navigation, clear dashboards, readable forms and consistent interface.
Security	Authentication, authorization, role-based access, secure password handling and protected data.
Reliability	Validation, error handling, database backups and recovery procedures.
Performance	Efficient database queries, optimized pages and appropriate server resources.
Scalability	Modular architecture and database design that can support more users, courses and departments.
Maintainability	Separate modules, clear documentation, coding standards and testable components.
Availability	Reliable hosting, monitoring, backup and recovery mechanisms.
Data Integrity	Validation, controlled updates, unique identifiers and database constraints.

4.5 Software Engineering Principles

Modularity

Each major function is separated into a module. Changes to assignments, schedules or announcements can therefore be made with limited impact on other modules.

Scalability

The design can start at department level and expand to multiple departments and larger user populations.

Maintainability

Structured modules, documentation and testing make defects easier to locate and features easier to update.

Reliability

Validation, exception handling, backups and recovery mechanisms reduce the effect of failures.

Usability

Role-specific dashboards and simple workflows reduce user confusion and training effort.

Security

Authentication, authorization, least-privilege access and secure data handling protect academic information.

4.6 Basic System Workflow

User Login → Role Verification → Role-Based Dashboard → Select Module → Perform Action → Validate Data → Store/Retrieve Data → Display Result → Generate Notification/Report where required.

4.7 Data and Security Considerations

- Student and faculty information should only be accessible to authorized roles.
- Assignment submissions and academic records should be protected from unauthorized modification.
- Regular database backups should be maintained.
- Input validation should reduce incorrect or harmful data.
- Important actions such as updates to schedules or records should be traceable through activity logs where required.

5. Methodology / Framework Justification

5.1 Selected Methodology – Agile with Scrum Practices

Agile is selected because the system has several independent but connected modules, and requirements can change after students, faculty and administrators review early versions. Agile allows the team to develop a working product in short iterations, collect feedback, test continuously and prioritize the most useful features.

5.2 Proposed Sprint Plan

Sprint	Main Deliverables
Sprint 1	Requirements, user roles, authentication and basic database structure.
Sprint 2	Course, classroom and timetable management.
Sprint 3	Course materials and assignment creation/submission.
Sprint 4	Announcements, notifications and student participation.
Sprint 5	Faculty activities, dashboards, reports, security and integration.
Sprint 6	System testing, user acceptance testing, bug fixing, deployment and documentation.

5.3 Why Agile is Suitable

21. Early working modules can be demonstrated to stakeholders.
22. Feedback can be incorporated before the complete system is built.
23. High-priority features can be developed first.
24. Testing can occur during each sprint rather than only at the end.
25. Changing academic requirements can be handled more easily.
26. Risks can be identified earlier through frequent reviews.

27. Continuous improvement supports long-term maintenance.

5.4 Maintenance Support

After deployment, new requirements can be added to the product backlog and developed in future iterations. Examples include mobile access, online examinations, attendance integration, AI-based analytics and additional reporting. This makes the system adaptable instead of requiring a complete redesign for every change.

6. Expected Outcomes and Limitations

6.1 Expected Benefits

28. Reduced manual work and paperwork.
29. Centralized and easier access to academic information.
30. Improved communication and timely announcements.
31. Better organization of course materials.
32. Simpler assignment submission and tracking.
33. Improved visibility of classroom schedules and conflicts.
34. More consistent student participation records.
35. Better monitoring of faculty activities.
36. Faster generation of reports and summaries.
37. Improved security through role-based access.

6.2 Challenge-to-Solution Mapping

Challenge	System Response	Expected Result
Scattered information	Centralized modules and database	Single, organized source of information
Manual assignment tracking	Assignment module	Faster submission and status tracking
Missed announcements	Announcements and notifications	Timely communication
Schedule conflicts	Schedule management	Better classroom/faculty allocation
Participation record difficulty	Participation module	Consistent digital records
Unauthorized access	Authentication and authorization	Improved data protection
Manual reporting	Dashboard and reporting	Faster academic monitoring

6.3 Potential Risks

- Cybersecurity attacks or unauthorized access.
- Server, database or network failure.
- Incorrect data entered by users.
- Data migration errors during initial deployment.
- Low user adoption or resistance to change.
- Performance problems when the number of users grows.
- Insufficient backup or recovery procedures.

6.4 Implementation Challenges

38. Training students, faculty and administrators.
39. Migrating existing timetable and academic records.
40. Integrating with existing institutional systems.
41. Providing reliable hosting and network infrastructure.
42. Maintaining privacy and security requirements.

43. Providing technical support after deployment.

6.5 Limitations

- Internet/network dependency may affect access in some deployments.
- Initial development, hosting and maintenance require resources.
- Users require basic digital skills.
- A system outage may temporarily interrupt access.
- The quality of reports depends on accurate data entry.

6.6 Future Enhancements

44. Android/iOS mobile application.
45. QR or biometric attendance integration.
46. Online examination and quiz module.
47. AI-based student performance analysis.
48. AI chatbot for common student queries.
49. Video classroom and live-session integration.
50. Email/SMS/push notifications.
51. Cloud deployment and automated scaling.
52. Advanced analytics and predictive dashboards.

Conclusion

The Smart Classroom Management System provides a centralized and structured solution for classroom schedules, course materials, assignments, announcements, student participation and faculty activities. The analysis shows that the traditional approach can create scattered information, manual effort, communication gaps and reporting difficulties.

The proposed modular solution addresses these gaps through role-based access, centralized data, dedicated functional modules, reporting and security controls. Applying modularity, scalability, maintainability, reliability, usability and security makes the proposed system practical and easier to evolve.

Agile with Scrum practices is appropriate because the system can be developed incrementally, reviewed frequently and improved using stakeholder feedback. The proposed system therefore provides a feasible foundation for improving classroom management while allowing future enhancements as institutional requirements change.